SKILL BUILDERS · UNIT 5 · SCORE 3

# Pythagorean Theorem: Practice Set

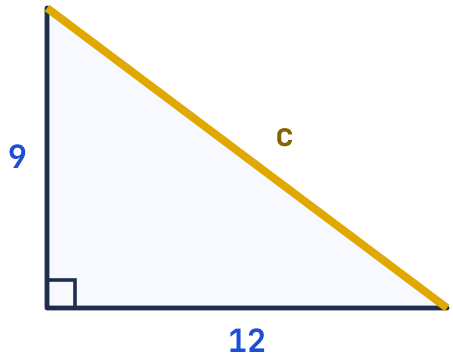
NC.8.G.6–8

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

**1** Score 3 NC.8.G.7

The legs of a right triangle are **9** and **12**. Find the hypotenuse.

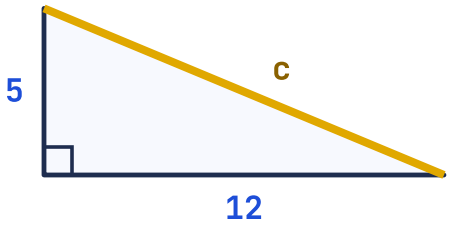



Answer: **15**

1. $9^2 + 12^2 = c^2$
2. $81 + 144 = c^2$
3. $c^2 = 225$
4. $c = \sqrt{225} = 15$

**2** Score 3 NC.8.G.7

The legs of a right triangle are **5** and **12**. Find the hypotenuse.

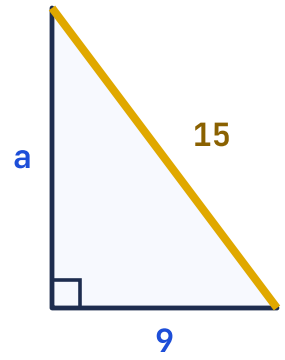



Answer: **13**

1. $5^2 + 12^2 = c^2$
2. $25 + 144 = c^2$
3. $c^2 = 169$
4. $c = \sqrt{169} = 13$

**3** Score 3 NC.8.G.7

A right triangle has hypotenuse **15** and one leg **9**. Find the other leg.

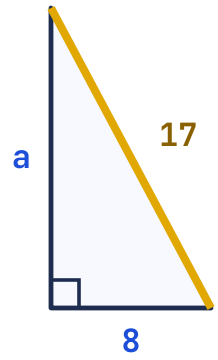



Answer: **12**

1. $a^2 + 9^2 = 15^2$
2. $a^2 = 225 - 81 = 144$
3. $a = \sqrt{144} = 12$

**4** Score 3 NC.8.G.7

A right triangle has hypotenuse **17** and one leg **8**. Find the other leg.




Answer: **15**

1. $a^2 + 8^2 = 17^2$
2. $a^2 = 289 - 64 = 225$
3. $a = \sqrt{225} = 15$

**5** Score 3 NC.8.G.6

Can the sides **7, 24, 25** make a right triangle?

(A) Yes: $49 + 576 = 625$, which is $25^2$

(B) No: the numbers are not 3, 4, 5

(C) No: $7 + 24$ is not 25

1. $7^2 + 24^2 = 49 + 576 = 625$
2. $25^2 = 625$
3. equal → right triangle

**6** Score 3 NC.8.G.6

Can the sides **4, 5, 6** make a right triangle?

(A) No: $16 + 25 = 41$, but $6^2 = 36$

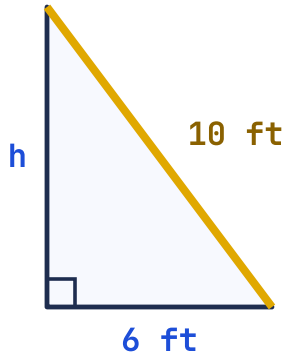
(B) Yes: $4 + 5$ is more than 6

(C) Yes: they are in counting order

1. $4^2 + 5^2 = 16 + 25 = 41$
2. $6^2 = 36$
3. not equal → not a right triangle

**7** Score 3 NC.8.G.7

A 10-foot ladder leans on a wall. The bottom is **6 feet** from the wall. How high up the wall does it reach?

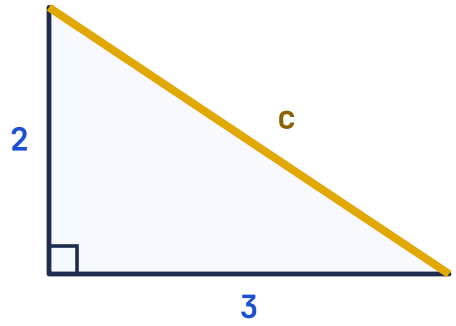



Answer: **8ft**

1. $a^2 + 6^2 = 10^2$
2. $a^2 = 100 - 36 = 64$
3. $a = \sqrt{64} = 8$

**8** Score 3 NC.8.G.7

The legs of a right triangle are **2** and **3**. Find the hypotenuse to the nearest tenth.




Answer: **3.6**

1. $2^2 + 3^2 = c^2$
2. $4 + 9 = c^2$
3. $c^2 = 13$
4. $c = \sqrt{13} \approx 3.6$

**9** Score 3 NC.8.G.8

Find the distance from **(−2, 1)** to **(2, 4)**.

Answer: **5**

1. run = 2 - (-2) = 4, rise = 4 - 1 = 3
2. $4^2 + 3^2 = 16 + 9 = 25$
3. distance = $\sqrt{25} = 5$

**10** Score 3 NC.8.G.7

**Find the mistake.** A right triangle has legs 6 and 8. Max wrote: $6 + 8 = 14$, so $c = 14$. What went wrong?

(A) Max is right: c = 14.

**(B) Max added the legs. He must square them: 36 + 64 = 100, so c = 10.**

(C) Max should subtract: 8 − 6 = 2.

(D) Max should multiply: 6 × 8 = 48.